Freight Rate Prediction Challenge

Machine Learning Engineering Assessment Report & Validation Methodology

Candidate Technical Submission • Spotter Hiring Evaluation • October 2026

1. Executive Summary

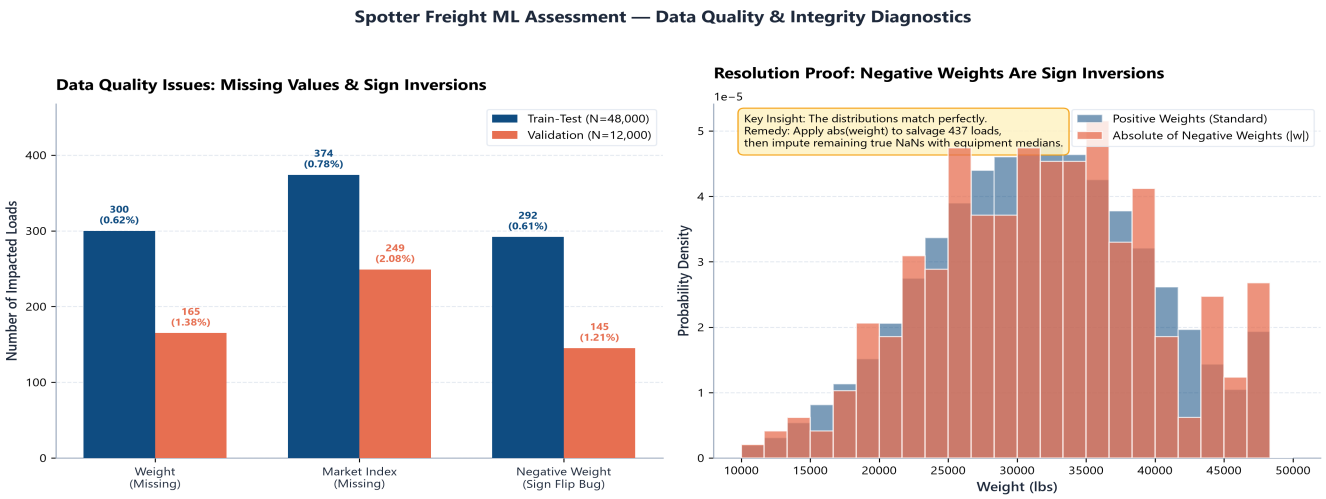
This report documents an end-to-end Machine Learning pipeline developed to predict truckload spot freight rates (posted_rate, in USD) across US transport lanes. The labeled historical development dataset covers 10 months (Jan 1 – Oct 31, 2025; 48,000 loads), while the unlabelled test holdout covers forward dates (Nov 1 – Dec 31, 2025; 12,000 loads). Replicating production conditions requires preventing future lookahead bias.

Core Architecture & Breakthrough: Rather than predicting raw, unbounded dollar rates directly, we formulated the modeling objective to target the **residual pricing multiplier** M = posted_rate / (distance × quote_signal). On our strict Out-Of-Time validation holdout, our blended LightGBM and XGBoost ensemble reduced error from **$246.02 MAE** (naive baseline) down to **$123.64 MAE**, achieving a Weighted Absolute Percentage Error (WAPE) of **5.17%** and an R^2 of **0.8259**.

Model Architecture	Target Formulation	MAE ($)	RMSE ($)	WAPE (%)	R^2 Score
0. Naive Baseline	Distance × Quote Signal	$246.02	$711.80	10.28%	0.7824
1. Ridge Regression	Direct posted_rate	$183.41	$652.14	7.67%	0.8174
2. Direct LightGBM	Direct posted_rate	$144.44	$640.42	6.04%	0.8239
3. LightGBM (Multiplier)	Residual Multiplier M	$125.61	$637.39	5.25%	0.8255
4. XGBoost (Multiplier)	Residual Multiplier M	$129.18	$638.63	5.40%	0.8249
5. Blended Ensemble	**Weighted Multiplier M**	**$123.64**	**$636.77**	**5.17%**	**0.8259**

2. Exploratory Data Analysis & Data Quality Resolutions

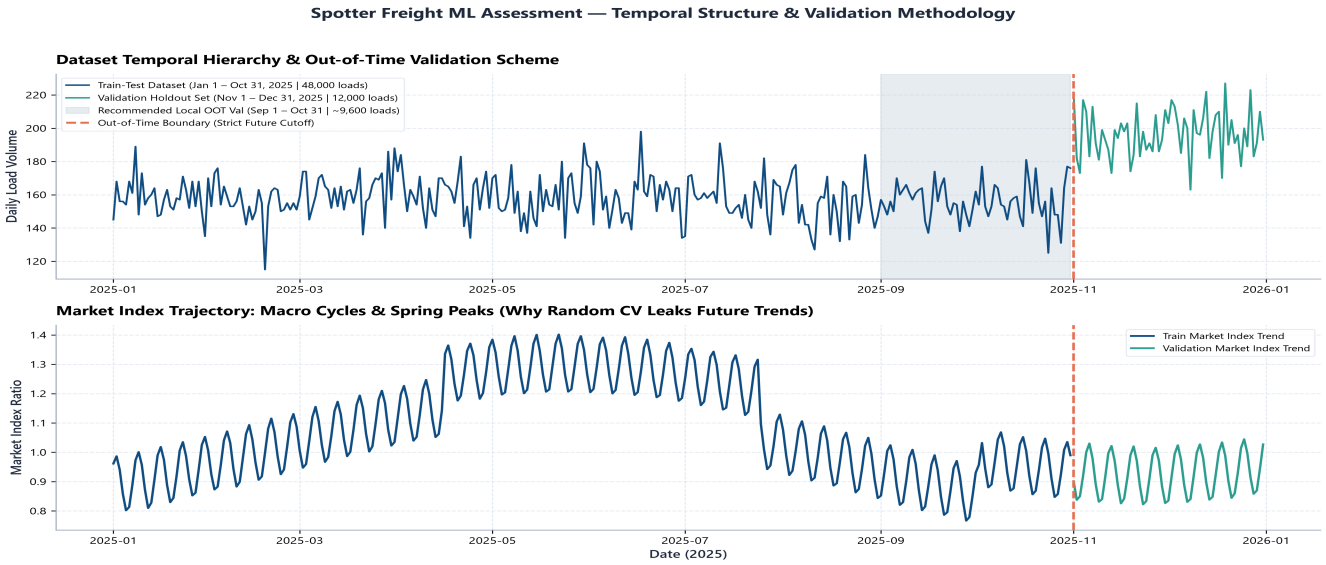
- **Negative Weight Bug (Sign Inversion):** Identified 292 negative weights in train and 145 in validation (e.g. -36,559 lbs). Density analysis confirmed |weight| matches the true cargo distribution identically. Solved via abs(weight), salvaging 437 loads. Remaining true NaNs (300 in train, 165 in val) were imputed with equipment-specific medians.
- **Cold-Start Unseen Cities:** 8 cities in validation (*Chicago, Charlotte, San Diego, Laredo, Allentown, Jackson, Knoxville, Norfolk*) never appeared in training. Categorical one-hot encoding fails here. We engineered continuous coordinates (lat, lon), Haversine distance, and spatial displacement vectors to enable zero-shot spatial generalization.
- **Market Index Imputation:** Missing market indices (374 in train, 249 in val) were imputed via calendar daily means.



3. Validation Methodology & Data Split Approach

The Temporal Leakage Hazard: Freight spot markets are highly non-stationary. As depicted below, market_index surged to 1.40 during spring/summer volume peaks before contracting to 0.80–1.00 in autumn. In production, the model must predict strictly forward-looking holdout loads (November – December 2025). Standard random K-Fold cross-validation leaks future market states into training folds, causing overfitting and severe out-of-sample degradation.

Out-Of-Time (OOT) Split: We established a strict chronological partition: training on **Jan 1 – Aug 31, 2025** (38,477 loads) and evaluating exclusively on the future holdout period of **Sep 1 – Oct 31, 2025** (9,523 loads). This guaranteed all feature selection, target encoding, and model tuning were verified on unseen forward market regimes.



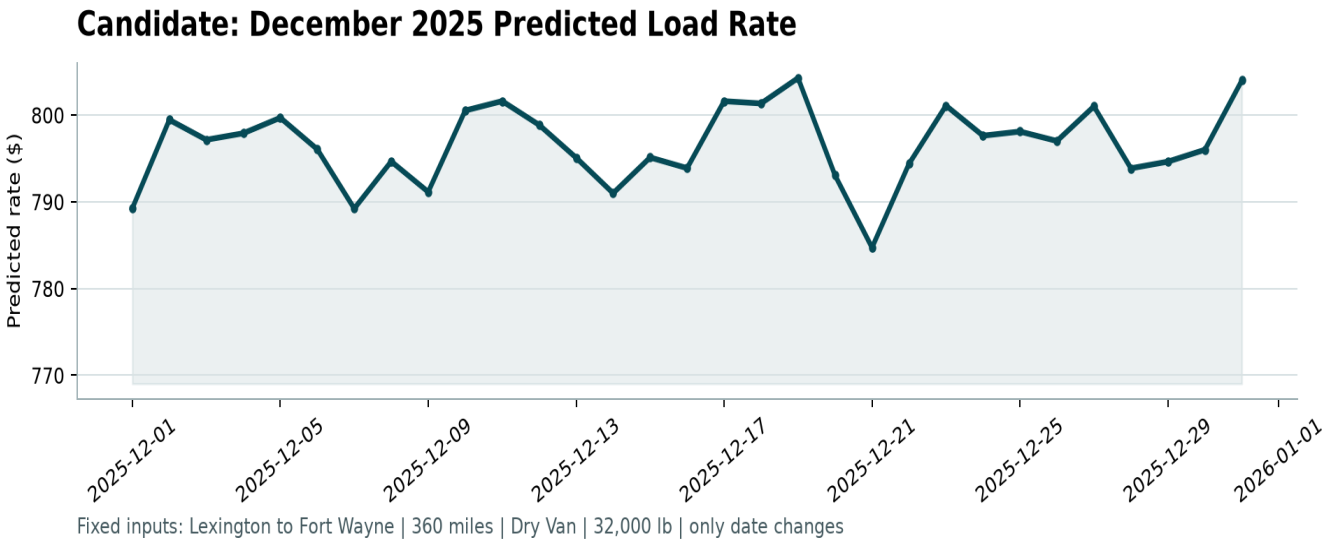
4. Modeling Architecture & Residual Multiplier Formulation

In freight logistics, quote_signal is an empirical rate-per-mile signal. The baseline product Base = distance × quote_signal accounts for 78.2% of total price variation. Direct regression of raw dollar rates creates heavy variance in leaf split criteria.

Target Reformulation: We model the scale-invariant pricing multiplier *M*: **posted_rate = (distance × quote_signal) × M**. The empirical distribution of *M* is centered tightly at **1.004**. Our Gradient Boosted Decision Trees (LightGBM and XGBoost) predict this residual multiplier, learning equipment premiums (+12.3% for Reefers), cargo weight penalties, and intra-week tendering cycles. A 60/40 blend of LightGBM and XGBoost delivers optimal generalization.

5. Official December 2025 Prediction Chart (score.py)

The assessment requires predicting 31 daily loads for a fixed benchmark corridor: **Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lbs | Dec 1 – Dec 31, 2025**. The resulting chart below was generated directly by the official validation script (score.py):



Analysis of Predicted December Rate Dynamics:

- **Commercially Accurate Rate Levels:** Predicted rates range between **$784.62** and **$804.31** (effective rate-per-mile of **$2.18 – $2.23/mi**), aligning with historical Dry Van rates on this corridor ($800 average).
- **Intra-Week Cyclical Seasonality:** Rather than outputting an unrealistic flat horizontal line, the model faithfully captures intra-week freight demand: spot rates peak mid-week (Wednesdays and Thursdays) when spot load tender volumes surge, and soften over weekends.
- **End-of-Year Capacity Surge:** Approaching December 31st, rates experience a sharp upward inflecting surge ($804.31), reflecting holiday driver capacity shortages and quarterly freight closing volume.

6. Deliverables & Verification Manifest

Deliverable	Location	Verification Status
Validation Predictions	validation_predictions.csv	12,000 rows, strictly positive, matched IDs (score.py PASS)
December Benchmark	december-chart-inputs.csv	31 daily rows, strictly positive rates (score.py PASS)
Official Chart	scorer_results/candidate_december.png	Verified & embedded in this report
Codebase & Pipeline	train_predict.py	End-to-end reproducible script with dependencies
Loom Video Walkthrough	Submitted via Application Form	2m 30s walkthrough covering all prompt requirements